energysafe VICTORIA
Creating a safer state with electricity and gas

2020 (2025)

# LICENSED ELECTRICIANS ASSESSMENT (LEA)

## Licensed Electricians Theory Examination
## SAMPLE EXAMINATION

**Surname:** ______________________________

**Given Names:** ______________________________

**Date:** ______________________________

- **PERSONAL NOTEPADS AND PAPER ARE NOT PERMITTED**
- **ONLY PENS MAY BE USED**
- **DO NOT REMOVE ANY SHEETS FROM THIS ASSESSMENT PAPER**
- **PAPERS WITH NO NAME OR SIGNATURE WILL NOT BE MARKED.**
- **UNITS MUST BE SHOWN TO OBTAIN FULL MARKS**

**The following reference books are permitted during the assessment session:**
AS/NZS 3000:2018 Wiring Rules (including amendments)
AS/NZS3012: 2019 Electrical Installations – Demolition and Construction Sites
AS/NZS 3008.1.1:2019 Electrical Installations – Selection of Cables
Electricity Safety (General) Regulations 2019
AS/NZS 4836:2023 Safe working on or near low voltage electrical installations and equipment

| Question | 1 | 2 | 3 | | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | TOTAL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Marks | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | |

Candidates need to obtain 75% or more to pass this assessment.

**Working Time: 2 hours**
**15 Minutes**

**At the end of this time you will be asked to stop.**

**Candidate Signature:** ______________________________

**Assessors Signature:** ______________________________

## WIRING RULES

In the **four** following Wiring Rules questions, you are required to:

- write the Wiring Rules Clause in the space provided
- the correct Wiring Rules Clause and Subclause must be given e.g. 3.5.2(b)(i)

**The correct answer to both parts must be given to obtain full marks.**

**Q 1.**
What word indicates a recommendation?

.......................................................................................................................................

Wiring Rules Clause number............................................................ [ 2 + 2 = 4 Marks]

**Q 2.**
Where wiring systems subject to vibration that is likely to cause damage to the wiring system, the wiring system shall be ……….

.......................................................................................................................................

.......................................................................................................................................

Wiring Rules Clause number................................................ [ 2 + 2 = 4 Marks]

**Q 3.**
Where a transformer is used to raise the voltage above that at which electricity is supplied, why would you make a connection between the primary and secondary windings with a protective earthing conductor?

.......................................................................................................................................

......................................................................................................................................

.......................................................................................................................................

Wiring Rules Clause number................................................

[ 2 + 2 = 4 Marks]

**Q 4.**
You have a power circuit with 5 sockets outlets connected, protected by a 20A C curve circuit breaker and run in 2.5mm$^2$, flat copper TPS. You are required to work out the voltage drop at the 2$^{nd}$ socket outlet using the formula Vd = L x I x Vc / 1000. State the value of the current you may use for I in the formula.

........................

Wiring Rules Clause number.................................................

[ 2 + 2 = 4 Marks]

## CONSTRUCTION & DEMOLITION SITES

In the following AS/NZS 3012 questions, you are required to:

- write the Standard's Clause number and/or Table number in the space provided
- the correct Clause and Sub-clause number must be given. e.g. 2.10.2 (f)

**The correct answer to both parts must be given to obtain full marks.**

**Q 5.**
In what condition is it permissible to install unarmoured cables on a metallic roof?

……………………………………………………………………………………………………

……………………………………………………………………………………………………

Standard Clause number…………………………….

[2 + 2 = 4 Marks]

**Q 6.**
How often should construction wiring be visually inspected to verify the integrity of the installation?

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

Standard Clause number............................................ [1+1+1+1= 4 Marks]

## ELECTRICITY SAFETY (GENERAL) REGULATIONS 2019

In the following Regulation question, you are required to:
* write your answers on the line/s below each question
* write the complete Regulation and Sub-Regulation number, if applicable, in the space provided. e.g. 401 (e) (3)

**The correct answer to parts and must be given to obtain full marks.**

**Q 7.**

As per the Electrical Safety (General) Regulations 2019. What is a voltage exceeding low voltage called?

....................................................................................................................................

**…………………………………………………………………………………………………..**

Regulation number.........................................................                    [2 + 2 = 4 Marks]

## ELECTRIC SHOCK SURVIVAL

**Q 8.**

In the absence of normal breathing state two actions you would take if no one has gone for help.

1. ..........................................................................................................................

2. ……………………………………………………………………………………………..

[2 + 2 = 4 Marks]

## CABLE SELECTION

**Q 9.**

**TWO** three-core X90 insulated, sheathed and armoured copper cables, including earthing conductors, are connected in parallel to supply a three-phase distribution board with a total maximum demand of 220A.

The cables are touching and protected by a circuit breaker and installed buried direct at a depth of 1.0m.

(i) Neglecting voltage drop, what is the cable size you would install for this circuit?
(ii) If using an HRC to protect this circuit, the current rating of the fuse should not exceed what value?

**Table details, calculations and units must be shown to obtain full marks.**

**Part (i)**

| | Answer | | Answer | | |
|---|---|---|---|---|---|
| Table 3 (______?) | | Item | | | |
| Table | | Column | | | Answer |
| Derate/rating table | | Column | | Factor | |
| Derate/rating table | | Column | | Factor | |

Part (i) A n s w e r ..................................................

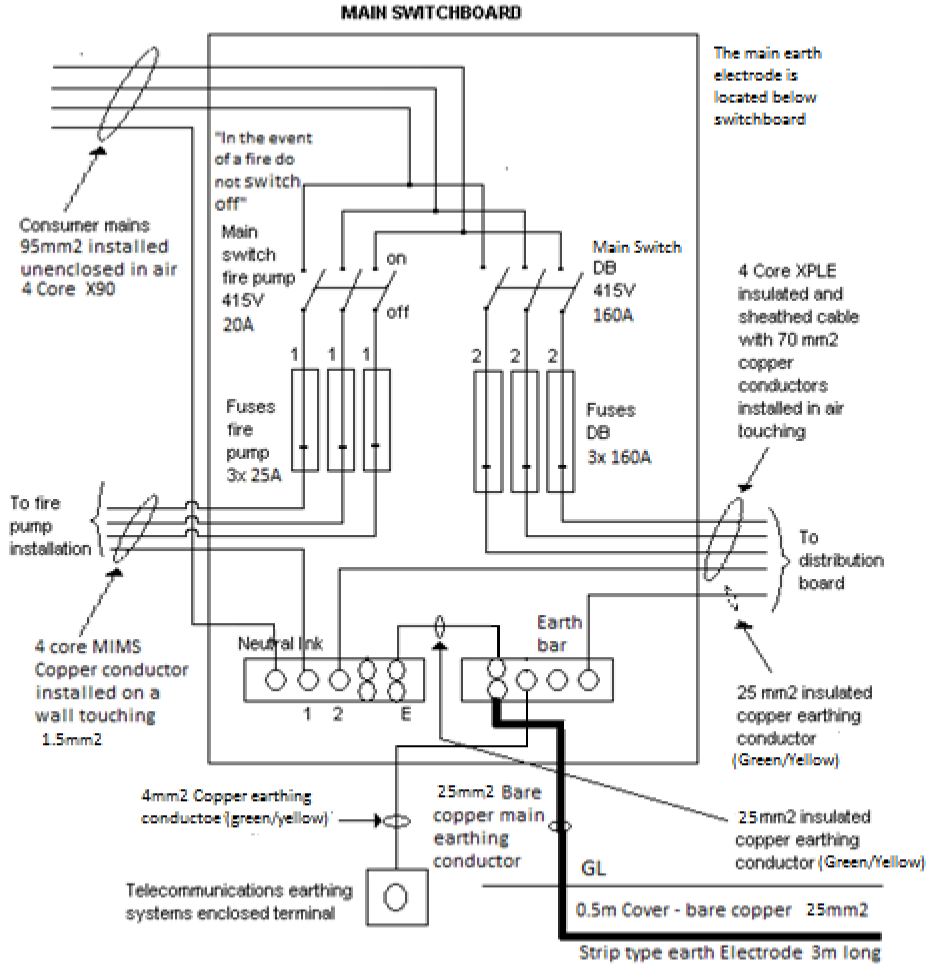
| | Answer | | Answer | | Answer |
|---|---|---|---|---|---|
| Derate/rating table | | Column | | Factor | |

Part (ii) Answer..................................................

]2+2+1+1+1+1 = 8 Marks]

## INSTALLATION DEFECTS - NON DOMESTIC

**Q 10.**




The drawing above shows the MAIN SWITCHBOARD of an industrial installation originating at the consumers mains and contains contraventions to the Wiring Rules.

It supplies a distribution board having a connected load with a calculated maximum demand of 160 A per phase and an automatically controlled 3 phase fire pump motor having a current rating of 20 A per phase.

The multi-core MIMS cables are installed spaced from the wall and are 1/1 kV cables

The supply authority has provided short circuit protection for the consumers mains.

The safety services main switch and the main switch for the general electrical installation are separated by a metal partition.

All screws in bars or links are all 80% of the tunnel diameter

Assume the MIMS cables are earthed in accordance with the Wiring Rules and are capable of maintaining supply to the equipment even when exposed to fire and mechanical damage.

All fuses shown are HRC type.

**Complete the table on the following page.**

**Q 10. continued.**

Use the diagram on the previous page.

List **FOUR different defects** together with the contravened Wiring Rules Clause/Table number in the table provided below.

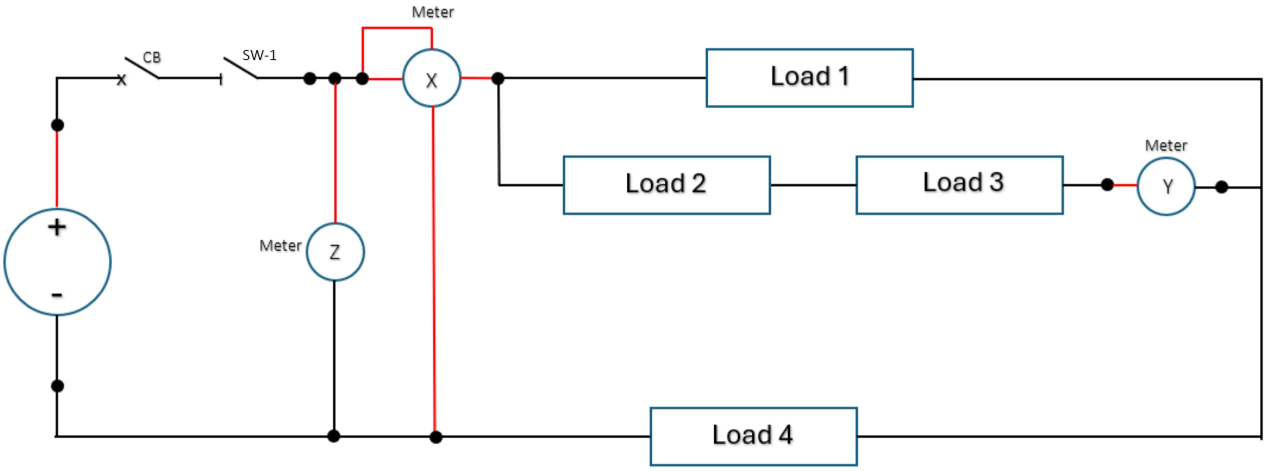
**Note: Only the first four defects will be considered**.

| DEFECT DETAILS | WIRING RULE CLAUSE/TABLE No. |
|---|---|
| | |
| | |
| | |
| | |

[ 4 x (2+1) = 12 Marks]

## DC CIRCUITS

**Q 11.**




For the circuit shown above, you have taken measurements and recorded the following:

Supply Current – 5A
Voltage across load 3 – 20V
Current through load 1 – 3A
Resistance of load 2 - 20Ω
Power dissipated by load 4 - 1900W

Using these measurements, calculate the meter readings for the following when the switch and circuit breaker is closed.

(i) Meter Y
(ii) Meter X
(iii) Meter Z

All calculations including the final answer must be completed to a maximum of **two decimal places**.

**Answers:** Meter X ........................ Meter Y …………………. Meter Z ........................

[3+3+3 = 9 Marks]

## MAXIMUM DEMAND

**Q 12.**

Calculate the Maximum Demand of a 230V sub-main supplying a distribution board in an air-conditioned boarding house.

The load connected to the switchboard is: -

- 3 circuits of 12 - 10A socket outlets
- 2 circuits of 2 - 20A socket outlets
- 3 circuits of 12 – incandescent lighting points.
- 1 – 8.2KW Range
- 1 – 6.2KW Split system air conditioner

**All relevant Table details, calculations and units must be shown to obtain full marks.**

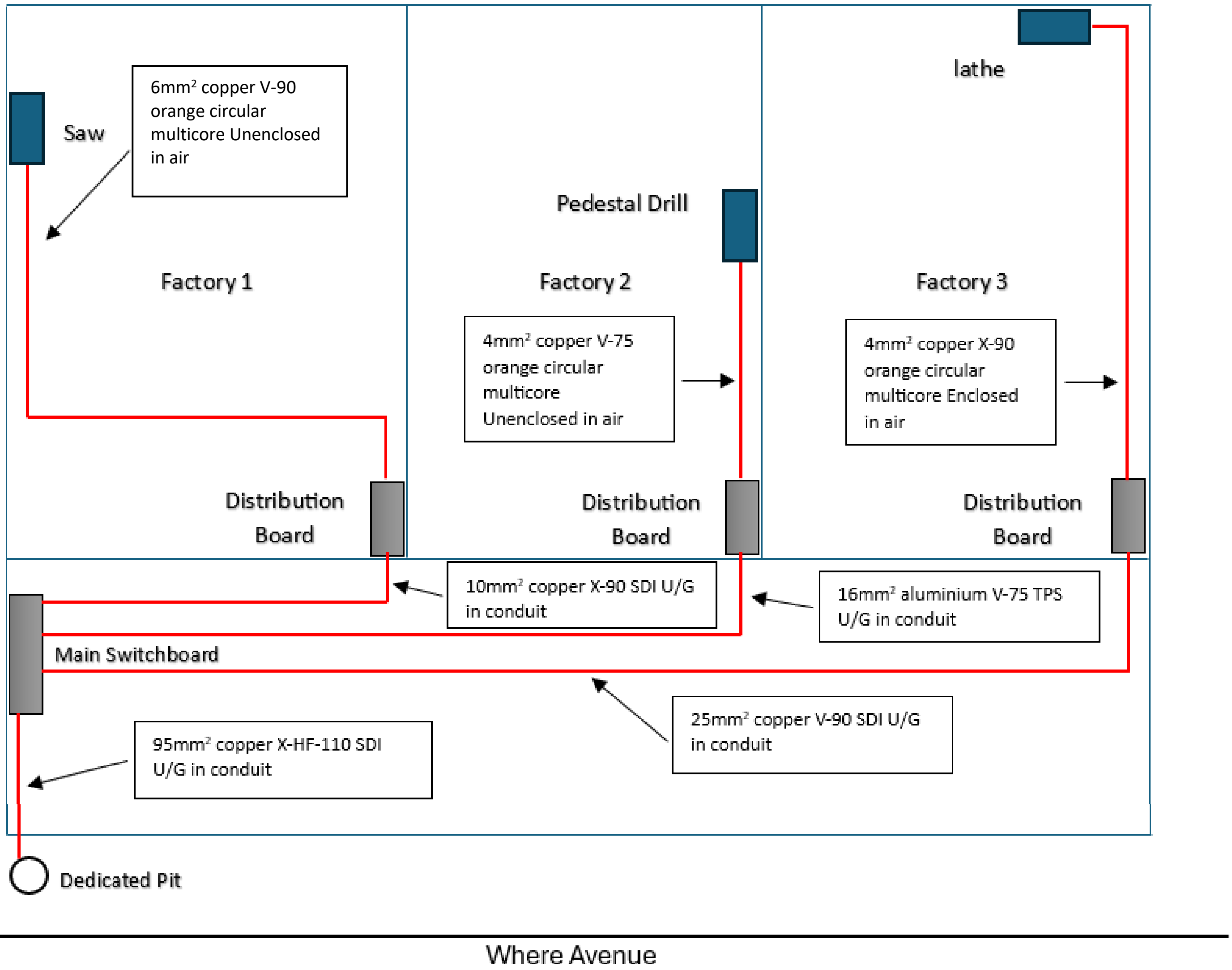
| Table | | Column | |
|---|---|---|---|
| | | | |
| Equipment | Load Group | Calculation | Maximum Demand |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | Total Maximum Demand | |

Answer.................................................

[1+2+2+2+1 = 8 Marks]

## VOLTAGE DROP

In a 400 /230 V, three-phase industrial installation, a three-phase 29A Saw is supplied from a sub-circuit originating at a distribution board in factory 1. The Saw is installed 32 meters away from the distribution board.




| **Client:** Ben Dover | **Location** | **Cable** | **Distance** | **Maximum Demand** | **Voltage** |
|---|---|---|---|---|---|
| **Drawing title:** Site plan | Main Switchboard | Consumers Mains | 10m | 125A | 400/230V |
| **Date:** 12 February 2025 | Factory 1 Distribution board | Sub Mains | 30m | 45A | 400/230V |
| **Location:** 2-5 Where Ave. North Haverbrook | Factory 2 Distribution board | Sub Mains | 70m | 50A | 400/230V |
| | Factory 3 Distribution board | Sub Mains | 110m | 55A | 400/230V |

Calculate the total voltage drop from the point of supply to the Saw terminals.

All calculations including the final answer must be completed to a maximum of **two decimal places**.

**All relevant table details, calculations and units must be show to obtain full marks**

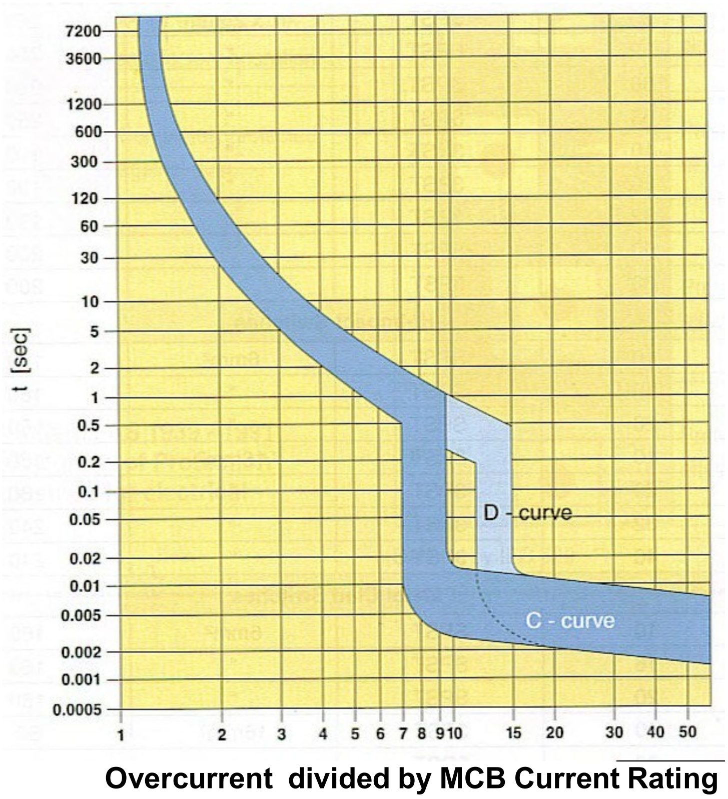
| Cable | Table | Column | Vc | Calculation | Vd |
|---|---|---|---|---|---|
| Consumer Mains | | | | | |
| Sub-mains | | | | | |
| Final Sub-circuit | | | | | |

| Answer Total Vd: | |
|---|---|

[3+3+3+1 = 10 marks]

## OVERLOAD & SHORT CIRCUIT CALCULATIONS

**Q 14.**
What are the minimum and maximum tripping times for a 63A Type C miniature over-current circuit breaker which is subjected to an over-current of 252A?




Overcurrent divided by MCB current rating………………

Answer: Minimum time......................... Maximum time...........................
[1+1+1 = 3 Marks]

## OVERLOAD & SHORT CIRCUIT CALCULATIONS

**Q 15.**
The main switchboard of a 400/230V industrial installation is directly supplied from a 500KVA transformer which has a prospective fault current of 13,990A per phase.

Submains supply a distribution board from the main switchboard.

The following information is known: -

Impedance of the Consumers Mains = 0.0080Ω
Impedance of the Submains cables = 0.059Ω

Determine the prospective fault current at: -

(i) Transformer impedance
(ii) the main switchboard; and
(iii) the distribution board.

Work impedances to 5 decimal places.

**All calculations must be shown to obtain full marks.**

Transformer Impedance ……….................................

Main SW/Bd ……………....................................

Dist/Bd …….........................................

[3+3+3 = 9 Marks]

## RESIDUAL CURRENT DEVICES

**Q 16.**
The load current rating of an RCD shall be not less than the greater of the following:

(a)……………………………………………………………………………………………………

……………………………………………………………………………………………………………

……………………………………………………………………………………………………

(b)……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

Standard Clause number.........................................

[3 Marks]

## MOTORS AND STARTERS

**Q 17.**

**CIRCLE** the letter in front of the statement you consider to be the most correct.

On start-up, a 3 phase 400V squirrel cage induction motor with an automatic star/delta starter:-

A starts on half the line voltage (200V).
B changes to delta connection at about 80% of the motor's full rated speed.
C has a starting torque three times that of the DOL starting torque.
D has a starting current one half of the DOL current.

[2 Marks]

## AS/NZS 4836:2011

**Q 18.**
This question relates to AS/NZS 4836:2011

During arc fault conditions, up to how many times the rated current of the supply transformer can fault current flow for short times?

.......................................................................................................................................

Standard Clause number......................................... [ 2 + 2 = 4 M